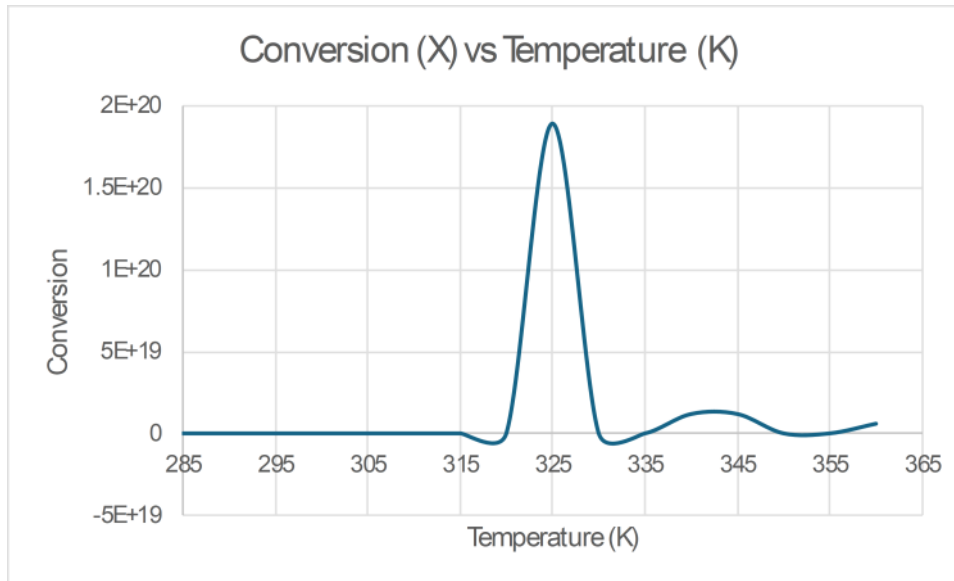


CHE 321 HW 4 Lindsey Meisch

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1. I would recommend slightly raising the reactor temperature from 300 K since the rate constant will increase with temperature.



The graph shows a maximum at 325 K, this is because of kinetics vs equilibrium. Raising the temperature increases conversion at first because of kinetics but then after the maximum, it drops because equilibrium limitation dominates.

2. $v_0 = 10 \text{ dm}^3/\text{min}$, $C_{A0} = C_{B0} = 1.0 \text{ M}$. Essentially constant v .
 $-r_A = k(C_A)(C_B) = k(C_{A0})^2(1-X)^2$
 $k(300\text{K}) = 0.070 \text{ dm}^3 \cdot \text{mol}^{-1} \cdot \text{min}^{-1}$
 $E = 20 \text{ kcal/mol} = 20,000 \text{ cal/mol}$
 $k(T) = k_{300} \cdot \exp(-(E/R)(1/T - 1/300))$
 $R = 1.987 \text{ cal} \cdot \text{mol}^{-1} \cdot \text{K}^{-1}$

 $0 \text{ C} \Rightarrow k(273 \text{ K}) = 2.535 \cdot 10^{-3}$
 $k(300 \text{ K}) = 0.070$
 $77 \text{ C} \Rightarrow k(350 \text{ K}) = 8.447$

- a) CSTR, 200 dm^3 , various T

$$F_{A0}X = V \cdot k \cdot (C_{A0})^2 \cdot (1-X)^2$$

$$a = \frac{V k C_{A0}^2}{F_{A0}} = \frac{2000 \text{ K}}{10} = 200 \text{ K}$$

$$X = a(1-X)^2 \Rightarrow aX^2 - (2a+1)X + a = 0$$

$$0 < X < 1$$

$$300 \text{ K} : a = 14 \Rightarrow X = 0.940$$

$$350 \text{ K} : a = 168.94 \Rightarrow X = 0.926$$

$$273 \text{ K} : a = 0.0507 \Rightarrow X = 0.046$$

PFT, 800 dm³ at 300 K

$$\frac{dX}{dV} = \frac{k C_{A0}^2 (1-X)^2}{F_{A0}} \Rightarrow \frac{1}{1-X} - 1$$

$$= \frac{k C_{A0}^2}{F_{A0}} V \equiv a \Rightarrow X = \frac{a}{1+a}$$

$$a = \frac{0.07}{10} 800 = 5.6 \Rightarrow X = 0.848$$

It's recommended to use the 200 dm³ CSTR heated to 77 C since it's gives the highest conversion of $X = 0.926$. The heating dominates, paint color is negligible.

$$b) \frac{dX}{dt} = k C_{A0} (1-X)^2$$

$$\Rightarrow t = \frac{1}{k C_{A0}} \left(\frac{1}{1-X} - 1 \right)$$

$$@ T = 350 K$$

$$k = 8.447$$

$$X = 0.90$$

$$t = \frac{1}{8.447} \left(\frac{1}{0.1} - 1 \right)$$

$$= \frac{9}{8.447} = \boxed{1.07 \text{ min}}$$

$$c) k(273 K) = 2.535 \times 10^{-3}$$

$$t = \frac{9}{0.002535}$$

$$= 3.55 \times 10^{-3} \text{ min}$$

$$= 59.2 \text{ h}$$

$$= 2.5 \text{ days}$$

$$d) \text{CSTR Series}$$

$$X_c = 0.440$$

$$C_{A,in} = C_{A0}(1-X_c)$$

$$X_c = 0.440$$

$$C_{A,in} = C_{A0}(1 - X_c)$$

$$C_{A,in} = 0.560$$

$$F_{A,in} = F_{A0}(1 - X_c)$$

$$F_{A,in} = 5.6037$$

PFR

$$a = \frac{0.07 C_{A,in}^2}{F_{A,in}} V$$

$$= 3.138$$

$$\Rightarrow X_{local} = \frac{a}{1+a}$$

$$= 0.758$$

Overall

$$X = 1 - (1 - 0.440)(1 - 0.758)$$

$$= 0.865$$

Parallel

CSTR

$$\frac{F_{A0}}{V} = 5$$

$$a = \frac{200(0.07)}{F_{A0}} = 2.8$$

$$\alpha = \frac{200(0.07)}{5} = 2.8$$

$$X_c' = 0.555$$

PFR

$$F_{A0} = 5$$

$$\alpha = \frac{0.07}{5} 800$$

$$= 11.2$$

$$X_p' = 0.918$$

$$X_{\text{parallel}} = \frac{1}{2}(0.555 + 0.918)$$

$$= 0.736$$

Series (0.865) > Parallel (0.736)

e) From part b,

$$t_{90} = \frac{9}{k_{300}}$$

$$= \frac{9}{0.07} = 128.6 \text{ min}$$

$$= 2.14 \text{ hr}$$

Batches per day

$$24/2.14 = 11.21$$

A moles fed per batch

$$C_{A0} V = 1 \times V$$

$$11.21 V = 14,400$$

$$\Rightarrow V = 1.28 \times 10^3 \text{ dm}^3$$

- f) I learned that being able to control the temperature dominates the performance of the reactor. Also, reactor staging matters (series gave higher conversion than parallel). For batch service, non reacting time is crucial for sizing. The point of this problem was to practice comparing reactor performances under different conditions.
- g) A "what-if" to consider is what if we ran the PFR at 77C? That might change which reactor is preferred.